

Power Series

Q. Find the radius and interval of convergence of power series:

1) $x - \frac{x^2}{3} + \frac{x^3}{5} - \frac{x^4}{7} + \dots$

$$U_n = \frac{(-1)^{n-1} x^n}{2n-1}$$

$$\therefore U_{n+1} = \frac{(-1)^{(n+1)-1} x^{n+1}}{2(n+1)-1}$$

$$= \frac{(-1)^n x^{n+1}}{2n+1} = \frac{(-1)^n x^n \cdot x}{2n+1}$$

* ↓ put modulus

$$\therefore \left| \frac{U_{n+1}}{U_n} \right| = \left| \frac{(-1)^n x^n \cdot x}{2n+1} \times \frac{2n-1}{(-1)^{n-1} x^n} \right|$$

$$= \left| \frac{(-1) \cdot (-1)^{n-1} x^n \cdot x}{2n+1} \times \frac{2n-1}{(-1)^{n-1} x^n} \right|$$

$$= \left| \frac{(-1) \cdot x \cdot (2n-1)}{2n+1} \right|$$

$$= |x| \times \frac{2n-1}{2n+1}$$

* For $n \geq 1$, $2n-1 > 0$

& $2n+1 > 0 \therefore \frac{2n-1}{2n+1} > 0$

$$= |x| \times \frac{n(2 - \frac{1}{n})}{n(2 + \frac{1}{n})}$$

$$= |x| \times \frac{(2 - \frac{1}{n})}{(2 + \frac{1}{n})}$$

$$\lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right| = \lim_{n \rightarrow \infty} |x| \times \frac{(2 - \frac{1}{n})}{(2 + \frac{1}{n})} = |x| \times \frac{2}{2} = |x|$$

* Imp^t
Mention
the test!!

By Ratio Test, the power series converges for $|x| < 1$

$$\hookrightarrow \lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right| = |x|$$

$\therefore$ Radius of convergence = 1

$\therefore$ Power Series converges for $-1 < x < 1$

$\hookrightarrow |x| < 1$ is same

$$\text{as } -1 < x < 1$$

* Now check the boundary points separately

Boundary points are $x = -1$ & $x = 1$

$$-1 < x < 1$$

i) For $x = -1$, $U_n = \frac{(-1)^{n-1} \cdot (-1)^n}{2^{n-1}} = \frac{(-1)^{2n-1}}{2^{n-1}} = \frac{-1}{2^{n-1}}$

$$\therefore \sum U_n = \sum \frac{(-1)}{2^{n-1}} = - \sum \frac{1}{2^{n-1}}$$

If +ve series converges/diverges, then correspondingly, the -ve series also cgs/diverges.

(say) $U_n = \frac{(-1)}{2^{n-1}} = \frac{1}{n(2 - \frac{1}{n})}$

$$\therefore V_n = \frac{1}{n}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{U_n}{V_n} = \lim_{n \rightarrow \infty} \frac{1}{2 - \frac{1}{n}} = \frac{1}{2} \neq 0$$

Now, $\sum V_n = \sum \frac{1}{n}$ is a p-series with $p=1$
 $\therefore \sum V_n$ diverges.

$\therefore$ By limit form of comparison test, $\sum U_n$ also diverges.
 $\therefore$ For $x = -1$, series diverges.

ii) For $x = 1$,

$$U_n = \frac{(-1)^{n-1} (1)^n}{2^{n-1}}$$

$\therefore$ The series is $\sum \frac{(-1)^{n-1}}{2^{n-1}}$ which is an

alternating series with $U_n = \frac{1}{2^{n-1}}$

$$\therefore U_{n-1} = \frac{1}{2^{(n-1)-1}} = \frac{1}{2^{n-2}}$$

$$\therefore U_n - U_{n-1} = \frac{1}{2^{n-1}} - \frac{1}{2^{n-2}} = \frac{2^{n-2} - (2^{n-1})}{(2^{n-1})(2^{n-2})}$$

$$= \frac{2^{n-2} - 2^{n-1}}{(2^{n-1})(2^{n-2})} = \frac{-2^{n-2}}{(2^{n-1})(2^{n-2})} = -2$$

$$\therefore U_n - U_{n-1} = \frac{-2}{(2^{n-1})(2^{n-2})} < 0 \text{ for } n > 1$$

* Imp't
Mention!!

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{1}{2n-1} = \lim_{n \rightarrow \infty} \frac{1}{n(2-\frac{1}{n})} = 0$$

∴ By Leibnitz's Test, series converges for $x=1$

* Series converges for $-1 < x < 1$ and $x=1$
series diverges for $x=-1$

∴ Interval of convergence is $-1 < x \leq 1$

less than equal to, because at $x=1$, series converges.

$$2) x + \frac{x^2}{8} + \frac{x^3}{27} + \frac{x^4}{64} + \dots$$

$$\rightarrow U_n = \frac{x^n}{n^3} \quad \therefore U_{n+1} = \frac{x^{n+1}}{(n+1)^3} = \frac{x^n \cdot x}{(n+1)^3}$$

$$\therefore \left| \frac{U_{n+1}}{U_n} \right| = \left| \frac{x^n \cdot x}{(n+1)^3} \times \frac{n^3}{x^n} \right| = \left| x \times \frac{n^3}{(n+1)^3} \right|$$

$$= |x| \times \left(\frac{n}{n+1} \right)^3 = |x| \times \frac{1}{\left(1 + \frac{1}{n}\right)^3}$$

$$\lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right| = \lim_{n \rightarrow \infty} |x| \times \frac{1}{\left(1 + \frac{1}{n}\right)^3}$$

$$= |x| \cdot \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^3} = \frac{|x|}{1} = |x|$$

∴ By Ratio Test, power series converges for $|x| < 1$
i.e., $-1 < x < 1$

∴ Radius of convergence = 1

i) For $x=-1$, the series is $\sum \frac{(-1)^n}{n^3}$ which is

alternating series with $U_n = \frac{1}{n^3}$ and $U_{n-1} = \frac{1}{(n-1)^3}$

$$\therefore U_n - U_{n-1} = \frac{1}{n^3} - \frac{1}{(n-1)^3} = \frac{(n-1)^3 - n^3}{n^3(n-1)^3}$$

$$= \frac{n^3 - 3n^2 + 3n - 1 - n^3}{n^3(n-1)^3}$$

$$U_n - U_{n-1} = \frac{-3n^2 + 3n - 1}{n^3(n-1)^3} < 0 \text{ for } n \geq 1$$

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{1}{n^3} = 0$$

$\therefore$ By Leibnitz's test, series converges for $x = -1$.

ii) For $x = 1$, series is $\sum \frac{1^n}{n^3} = \sum \frac{1}{n^3}$ which

is a p-series with $p = 3 > 1$

$\therefore$ Series converges for $x = 1$

$\therefore$ Interval of convergence is $-1 \leq x \leq 1$

Both less than equal to
because series cgs at $x = -1 \neq$

$x = 1$ also

$$3) \sum_{n=1}^{\infty} \frac{x^n}{3^n n^2}$$

$$\rightarrow U_n = \frac{x^n}{3^n n^2}$$

$$U_{n+1} = \frac{x^{n+1}}{3^{n+1} (n+1)^2}$$

$$\left| \frac{U_{n+1}}{U_n} \right| = \left| \frac{x^{n+1}}{3^{n+1} (n+1)^2} \times \frac{3^n n^2}{x^n} \right|$$

$$= \left| \frac{x^n \times x}{3^n \times 3 \times (n+1)^2} \times \frac{3^n \times n^2}{x^n} \right|$$

$$= \left| \frac{x}{3} \times \frac{n^2}{(n+1)^2} \right| = \frac{|x|}{3} \times \left(\frac{n}{n+1} \right)^2 = \frac{|x|}{3} \times \frac{1}{\left(1 + \frac{1}{n}\right)^2}$$

$$\therefore \lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right| = \lim_{n \rightarrow \infty} \frac{|x|}{3} \times \frac{1}{\left(1 + \frac{1}{n}\right)^2} = \frac{|x|}{3}$$

By Ratio test, power series converges

for $\frac{|x|}{3} < 1$ i.e. $|x| < 3$

i.e. $-3 < x < 3$

$$|x| < 3$$

$\therefore$ Radius of convergence = 3

i) For $x = -3$, series is $\sum \frac{(-3)^n}{3^n n^2}$

$$= \sum \frac{(-1)^n 3^n}{3^n n^2} = \sum \frac{(-1)^n}{n^2}$$

which is an alternating series with

$$U_n = \frac{1}{n^2} \quad \therefore U_{n-1} = \frac{1}{(n-1)^2}$$

$$\begin{aligned} \therefore U_n - U_{n-1} &= \frac{1}{n^2} - \frac{1}{(n-1)^2} = \frac{(n-1)^2 - n^2}{n^2(n-1)^2} \\ &= \frac{n^2 - 2n + 1 - n^2}{n^2(n-1)^2} \\ &= \frac{-2n + 1}{n^2(n-1)^2} < 0 \quad \text{for } n > 1 \end{aligned}$$

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{1}{n^2} = 0$$

$\therefore$ By Leibnitz's test, series converges for $x = -3$

ii) For $x = 3$, series is $\sum \frac{3^n}{3^n n^2} = \sum \frac{1}{n^2}$ which

is a p-series with $p = 2 > 1$

$\therefore$ Series converges for $x = 3$.

$\therefore$ Interval of convergence is $-3 \leq x \leq 3$.

4) $\frac{x}{\sqrt{1}+1} - \frac{x^2}{\sqrt{2}+1} + \frac{x^3}{\sqrt{3}+1} - \frac{x^4}{\sqrt{4}+1} + \dots$

$$\rightarrow U_n = \frac{(-1)^{n-1} x^n}{\sqrt{n} + 1} \quad \therefore U_{n+1} = \frac{(-1)^n x^{n+1}}{\sqrt{n+1} + 1}$$

$$\begin{aligned} \left| \frac{U_{n+1}}{U_n} \right| &= \left| \frac{(-1)^n x^{n+1}}{\sqrt{n+1} + 1} \times \frac{\sqrt{n} + 1}{(-1)^{n-1} x^n} \right| \\ &= \left| (-1) x \times \frac{\sqrt{n} + 1}{\sqrt{n+1} + 1} \right| = |x| \times \frac{\sqrt{n} + 1}{\sqrt{n+1} + 1} \end{aligned}$$

$$\frac{\sqrt{n+1}}{\sqrt{n+1}+1} = \frac{\sqrt{n} \left(1 + \frac{1}{\sqrt{n}}\right)}{\sqrt{n} \left(1 + \frac{1}{\sqrt{n}}\right) + 1}$$

$$\frac{\sqrt{n} \left(1 + \frac{1}{\sqrt{n}}\right)}{\sqrt{n} \left(1 + \frac{1}{\sqrt{n}}\right) + 1}$$

$$= \frac{\sqrt{n} \left(1 + \frac{1}{\sqrt{n}}\right)}{\sqrt{n} \left(1 + \frac{1}{\sqrt{n}}\right) + 1}$$

$$\sqrt{n} \left[\sqrt{1 + \frac{1}{n}} + \frac{1}{\sqrt{n}} \right]$$

$$\frac{\sqrt{n+1}}{\sqrt{n+1}+1} = \frac{1 + \frac{1}{\sqrt{n}}}{\sqrt{1 + \frac{1}{n}} + \frac{1}{\sqrt{n}}}$$

$$\therefore \left| \frac{U_{n+1}}{U_n} \right| = |x| \times \frac{1 + \frac{1}{\sqrt{n}}}{\sqrt{1 + \frac{1}{n}} + \frac{1}{\sqrt{n}}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right| = \lim_{n \rightarrow \infty} |x| \times \frac{1 + \frac{1}{\sqrt{n}}}{\sqrt{1 + \frac{1}{n}} + \frac{1}{\sqrt{n}}} = |x| \times \frac{1+0}{\sqrt{1+0}+0}$$

$$= |x|$$

$\therefore$ By Ratio Test, power series converges for $|x| < 1$
ie $-1 < x < 1$

Radius = 1.

i) For $x = -1$, the series is $\sum \frac{(-1)^{n-1} (-1)^n}{\sqrt{n+1}}$

$$= \sum \frac{(-1)^{2n-1}}{\sqrt{n+1}} = - \sum \frac{1}{\sqrt{n+1}}$$

$$U_n = \frac{1}{\sqrt{n+1}} = \frac{1}{\sqrt{n} \left(1 + \frac{1}{\sqrt{n}}\right)}$$

$$V_n = \frac{1}{\sqrt{n}} = \frac{1}{n^{1/2}}$$

$$\lim_{n \rightarrow \infty} \frac{U_n}{V_n} = \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{\sqrt{n}}} = 1 \neq 0$$

Also, $\sum V_n = \sum \frac{1}{n^{1/2}}$ is a p-series with

$p = \frac{1}{2} < 1 \therefore \sum V_n$ diverges

$\therefore$ By comparison test, $\sum U_n$ also diverges

$\therefore$ Series diverges for $x = -1$.

ii) For $x = 1$, series is $\sum \frac{(-1)^{n-1} (1)^n}{\sqrt{n} + 1} = \sum \frac{(-1)^{n-1}}{\sqrt{n} + 1}$

which is an alternating series with

$$U_n = \frac{1}{\sqrt{n} + 1} \therefore U_{n-1} = \frac{1}{\sqrt{n-1} + 1}$$

$$\text{Now, } n-1 < n$$

$$\therefore \sqrt{n-1} < \sqrt{n}$$

$$\therefore \sqrt{n-1} + 1 < \sqrt{n} + 1$$

$$\Rightarrow \frac{1}{\sqrt{n-1} + 1} > \frac{1}{\sqrt{n} + 1}$$

$$\text{ie } U_{n-1} > U_n$$

$$\therefore U_n - U_{n-1} < 0 \text{ for } n > 1.$$

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n} + 1} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n} \left(1 + \frac{1}{\sqrt{n}}\right)} = 0$$

$\therefore$ By Leibnitz Test, series converges at $x = 1$.

$\therefore$ Interval of convergence is $-1 < x \leq 1$.